Robust Regression under Distributional Contamination: A Monte Carlo Study of OLS, LTS, Theil and Bayesian Estimators

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ABSTRACT
The Ordinary Least Squares (OLS) estimator, while standard for linear regression, is highly susceptible to outliers and heavy-tailed error distributions due to its reliance on normality and homoscedasticity. This study utilizes the ADEMP and Tidy Simulation frameworks to conduct an extensive Monte-Carlo simulation (150,000 replications) to evaluate classical, robust, and Bayesian estimators within simple linear regression. We performed simulations across six sample sizes and four error-generating environments (Standard, Contamination, Outlier, and Mixture models). Estimators - OLS, Least Trimmed Squares (LTS), Theil's pair-wise median, and Bayesian regression - were compared using bias, mean square error (MSE), Monte Carlo standard error (MCSE), relative mean square error (RMSE), and interval coverage probability. Furthermore, kernel density estimation was employed to assess the empirical behaviour, structural stability, and convergence characteristics of the simulated error distributions. Our results demonstrate that Theil's method consistently exhibits the highest reliability across all conditions; however, its computational complexity restricts its use in large-scale datasets. The LTS method offers the optimal balance between computational efficiency and robustness for larger samples. Conversely, standard Bayesian regression, when implemented without robust heavy-tailed priors, remains as vulnerable to outliers as OLS. These findings provide empirical guidance for selecting robust statistical frameworks, confirming that transitioning from OLS is essential when data deviates from Gaussian assumptions.

Keywords: Robust Statistics; Monte Carlo Simulation; Breakdown Point; Estimator Efficiency, Outlier Resistance, Heavy-Tailed Distributions, Simple Linear Regression.

Declarations

Disclosure Statement

The author reports there are no competing interests to declare.

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Data Availability Statement

The underlying simulation code developed for this research is publicly accessible GitHub. The repository includes all Python scripts used for the simulation and model estimation, simulation results table for n = 30, 50 and 500, simulated replication data and GDP and Population data for empirical analysis.

1.0 INTRODUCTION

The frequentist perspective of linear regression represents the most conventional approach to statistical modelling. It provides parameter estimates based on the foundational assumption that the model is fully informed by the provided dataset alone. However, classical frequentist procedures, most notably Ordinary Least Square Estimation (OLS), are known to breakdown under small sample conditions. Furthermore, OLS is highly susceptible to outliers; research has demonstrated that even a single outlier within a sample can significantly skew regression estimates. Consequently, researchers have developed alternative methods that offer considerable advantages over the OLS procedure and provide greater practical utility.

This study evaluates three specific alternatives to OLS: the Least Trimmed Squares (LTS) estimator, introduced by Rousseeuw in 1984; Theil's pairwise median procedure, introduced in 1950, which is expected to perform effectively regardless of the distribution of the error terms; and the Bayesian inference method, which traces its origins to Thomas Bayes in 1763 and was subsequently expanded by Pierre-Simon Laplace and others.

2.0 LITERATURE REVIEW

2.1 Overview of Regression Robustness

The Ordinary Least Squares (OLS) estimator has long served as the cornerstone of linear regression analysis. Historically, its dominance is attributed to the Gauss-Markov theorem, which establishes that under ideal conditions, where errors are normally distributed, homoscedastic, and uncorrelated, OLS provides the Best Linear Unbiased Estimator (BLUE). However, the practical application of OLS is frequently undermined by the realities of messy data. As observed by Birkes and Dodge (1993), while OLS is mathematically elegant, its performance degrades rapidly when these restrictive assumptions are violated.

A primary challenge in classical regression is its sensitivity to outliers. Because OLS minimizes the sum of squared residuals, it assigns disproportionate weight to these extreme values, often skewing the intercept and slope coefficients to such a degree that the resulting model loses its predictive and interpretive validity (Dietz, 1987; Rousseeuw & Leroy, 1987). This vulnerability is formally quantified by the "breakdown point", a metric developed to assess an estimator's resistance to contamination. The breakdown point represents the proportion of incorrect observations (outliers) an estimator can handle before yielding arbitrarily large or erroneous results (Hampel, 1974). Classical OLS is characterized by a breakdown point of $\frac{1}{n}$, which approaches 0% as the sample size n increases, rendering the estimator essentially defenseless against even a single influential outlier (Rousseeuw & Yohai, 1984).

The limitations of OLS are particularly pronounced when error terms deviate from normality, such as in the presence of heavy-tailed distributions or mixtures. In such scenarios, the reliance on squared residuals not only reduces the efficiency of the estimates but can also lead to misleading statistical inferences (Agullo, 2001; Hussain & Sprent, 1983).

Consequently, there has been a significant shift toward robust statistical frameworks capable of maintaining stability despite data contamination. Robust regression methods are designed to limit the influence of outliers while preserving the integrity of the underlying model. By employing alternative loss functions, such as those used in Least Trimmed Squares (LTS) or Theil's pairwise median procedure, these methods offer a safeguard against the breakdown that plagues classical approaches (Iman & Conover, 1979; Theil, 1950). Modern research continues to emphasize that the transition to robust methods is not merely an optional optimization but a necessary evolution in data analysis, particularly as practitioners encounter increasingly complex, high-dimensional, and noisy datasets where traditional OLS assumptions are rarely met (Morris et al., 2019; Sami, 2023).

2.2 The Rise of Robust Methods

As the limitations of Ordinary Least Squares (OLS) became increasingly apparent in the presence of non-Gaussian error terms and outliers, the statistical community shifted toward developing non-parametric and robust estimation frameworks. These methods were engineered to offer resistance to contamination while maintaining reliable parameter estimates where OLS fails. Two of the most significant advancements in this field are Theil's pairwise median procedure and the Least Trimmed Squares (LTS) estimator.

Theil's method, introduced by Theil (1950), is a rank-invariant, non-parametric approach that relies on the median of slopes between all pairs of observations. Because it is based on the median rather than the mean, it is inherently resistant to outliers, as it does not assign excessive weight to extreme deviations (Theil, 1950). This rank-based approach serves as a robust alternative to OLS, effectively bypassing the requirement for normally distributed error terms. Recent literature confirms that such non-parametric methods remain essential in modern statistical applications, providing stable inferences even when data is heavily contaminated (Hussain & Sprent, 1983; Sami, 2023).

Complementing this, the Least Trimmed Squares (LTS) estimator emerged as a powerful solution for robust regression, formally introduced by Rousseeuw (1984). LTS operates by minimizing the sum of the smallest squared residuals, thereby effectively "trimming" the impact of influential outliers that would otherwise dominate the OLS criterion (Rousseeuw & Leroy, 1987). Unlike OLS, which possesses a breakdown point of $\frac{1}{n}$, LTS offers a significantly higher breakdown point, providing superior resilience in the presence of bad data points (Agullo, 2001).

While these methods were computationally prohibitive at the time of their inception, modern advancements in Markov Chain Monte Carlo (MCMC) simulations and improved algorithms have revitalized their utility (Sami, 2023). Today, these robust techniques are widely validated in comparative studies, where they consistently outperform classical estimators in scenarios involving heavy-tailed distributions and heteroscedasticity (Ibrahim et al., 2022; Rather et al., 2022). Consequently, the integration of LTS and Theil's method represents a critical evolution in statistical practice, ensuring that empirical models remain grounded in reality despite the prevalence of noise and outliers in complex datasets.

2.3 Bayesian Integration

The transition of Bayesian methods from theoretical constructs to practical statistical tools has been one of the most significant developments in modern data science. Historically, the Bayesian perspective was limited by analytical intractability; however, the advent and refinement of Markov Chain Monte Carlo (MCMC) methods have allowed practitioners to estimate complex posterior distributions that were previously impossible to compute (Bolstad, 2010; Gelman et al., 2013). This computational revolution has positioned Bayesian inference as a flexible alternative to frequentist procedures, offering a framework that inherently accounts for uncertainty through the integration of prior information with observed data.

Despite this flexibility, the performance of Bayesian models relative to robust frequentist estimators, such as Theil's method and Least Trimmed Squares (LTS), remains subject to significant ambiguity. A critical misconception in the literature is the assumption that Bayesian regression is inherently robust. In practice, standard Bayesian regression utilizing conjugate Normal priors mirrors the limitations of Ordinary Least Squares (OLS), remaining highly sensitive to outliers and heavy-tailed error distributions (Giacomini et al., 2021). To achieve true robustness, Bayesian models must employ heavy-tailed likelihoods (e.g., Student-t distributions) or robust priors, yet the comparative efficacy of these "robust Bayesian" configurations against non-parametric methods like Theil's remains under-explored in the existing literature (Guerard et al., 2021; Zhang et al., 2021).

There is a distinct research gap regarding direct, head-to-head simulation comparisons of these disparate paradigms. While robust frequentist estimators have been extensively benchmarked for their breakdown points and computational efficiencies, Bayesian methods are often evaluated in isolation or strictly within a purely Bayesian context. Furthermore, many existing studies fail to test these methods under standardized, rigorous frameworks, such as the ADEMP (Aims, Data-generating mechanisms, Estimands, Methods, and Performance measures) guidelines, which makes cross-method comparisons unreliable (Morris et al., 2019). Consequently, there is an urgent need to empirically evaluate how Bayesian methods, when configured for robustness, perform against non-parametric alternatives across varying sample sizes and levels of data contamination. This study addresses this gap by subjecting both paradigms to the same rigorous Monte Carlo simulation environment, providing a neutral ground for evaluating their respective stability and performance.

2.4 State of Simulation Methodology

The empirical evaluation of statistical estimators is fundamentally a trade-off between statistical efficiency (the precision and unbiasedness of the estimate) and computational efficiency (the time and hardware resources required for computation). While Theil's pairwise median estimator is widely lauded for its robustness and rank-invariance, its computational complexity of $O(n^2)$ presents significant challenges for large-scale datasets, often restricting its utility in real-time or massive data environments (Theil, 1950; Sami, 2023). Conversely, the Least Trimmed Squares (LTS) estimator provides a superior breakdown point, yet its original implementation was notoriously difficult to compute. Advancements in algorithmic efficiency, such as the methodology introduced by Agullo (2001), have mitigated these barriers, allowing LTS to serve as a more scalable robust alternative to Theil's method in modern applications.

The selection of these three specific estimators, Theil’s method, LTS, and Bayesian inference, for this research is intended to cover the three pillars of robust statistical modelling. Theil’s method provides a benchmark for classic non-parametric robustness; LTS offers high-breakdown resistance to outliers; and Bayesian inference introduces the modern standard for uncertainty quantification through MCMC methods (Gelman et al., 2013). By evaluating these three paradigms head-to-head, this study provides a comprehensive assessment that bridges the gap between traditional frequentist robustness and contemporary probabilistic inference.

Crucially, the integrity of this evaluation relies on the reproducibility and design of the simulation itself. The methodology of this research is grounded in the ADEMP framework (Aims, Data-generating mechanisms, Estimands, Methods, and Performance measures), which ensures that the simulation process is rigorously defined and transparent (Morris et al., 2019). Furthermore, this study utilizes the "Tidy Simulation" framework, which facilitates the design of scalable and reproducible Monte Carlo experiments (van Kesteren, 2026). While previous studies have compared these estimators, many have failed to incorporate modern simulation frameworks like ADEMP, which this study addresses. By adopting these rigorous methodological standards, this research aims to provide a definitive resolution to the on-going debate regarding the optimal performance of these estimators under non-normal and contaminated data conditions.

3.0 METHODOLOGY

3.1.0 Simulation Design

This study employs the ADEMP framework (Aims, Data-generating mechanisms, Estimands, Methods, and Performance measures) to conduct a rigorous Monte Carlo simulation. By utilizing the “Tidy Simulation” paradigm, we ensure the experiment is scalable, reproducible, and provides a controlled environment to compare the performance of four regression estimators across varying levels of data contamination.

3.1.1 Data-Generating Mechanism (DGM)

The simulation considers a simple linear regression model:

$$Y_t = \alpha + \beta X_t + \varepsilon_t \text{ for } \alpha = 5 \text{ and } \beta = 2$$

where the design variable X is generated as a sequential model of the form $X_t = t;\ t = 1,2,3, \dots, n$, and ε_t is the error term. To evaluate estimator performance under non-ideal conditions, we generated errors from three distinct distributions:

1. **Normal:** $N(0, 1)$ (Baseline).
2. **Lognormal:** L(0,1) Skewed, non-symmetric distribution.
3. **Cauchy:** C(0,1) Heavy-tailed distribution with undefined mean and variance.

Furthermore, we introduced three contamination scenarios: Outliers (extreme values injected into Y), Mixtures (a weighted combination of two distributions), and Contaminations (data corruption). These scenarios were tested across six sample sizes $(n = 10, 30, 50, 100, 500, 1000)$ with $N = 150{,}000$ iterations per condition to ensure stability in the performance metrics.

For this study, the alternative error distributions were constructed as:

$$(1 - \epsilon)F(x) + \epsilon G(x)$$

where $F(x)$ represents the standard error distribution, $G(x)$ denotes the contaminating error distribution and ϵ is the probability of contamination.

The empirical behaviours of the simulated error data was characterized using Kernel Density Estimation (KDE) plots across the six sample sizes and under the four distinct structural frameworks considered in the study. Within each model, the three error distributions under investigation (Normal, Lognormal, and Cauchy) were assessed in order to evaluate structural stability, tail behaviour, and asymptotic convergence properties under varying degrees of stochastic perturbation.

3.1.2 Choice of Contaminating Distribution

Unlike the symmetric Normal and Cauchy distributions, or the smoothly continuous Lognormal distribution, the geometric distribution, G(p), is a discrete model characterized by heavy right-skewness across its non-negative support. Because of these distinct properties, the geometric distribution is frequently utilized as a destructive contaminant. It undermines the integrity of statistical models by forcing discrete characteristics onto continuous data, thereby destroying symmetry, introducing severe directional bias, and significantly impairing the effectiveness of robust estimation methods.

Table 1: Alternative Error Distribution Models

$\epsilon = 0.2$ Seed = n	STANDARD MODEL	OUTLIERS	MIXTURES	CONTAMINATIONS
NORMAL	N(0,1)	N(0,10)	N(10,10)	G(0.5)
LOG-NORMAL	LN(0,1)	LN(0,10)	LN(10,10)	G(0.5)
CAUCHY	C(0,1)	C(0,10)	C(10,10)	G(0.5)

3.2 Mathematical Specification of Estimators

The choice of estimators represents a trade-off between statistical efficiency and robustness. The objective functions for each method are specified below:

- Ordinary Least Squares (OLS): This estimator minimizes the sum of squared residuals:

$$\min_{\beta} \sum_{i=1}^{n} \left(y_i - (\alpha + \beta x_i)\right)^2$$

OLS provides the Best Linear Unbiased Estimator (BLUE) under Gaussian assumptions but possesses a breakdown point of $\frac{1}{n}$, rendering it highly sensitive to outliers.

- Least Trimmed Squares (LTS): LTS improves robustness by minimizing the sum of the h smallest squared residuals:

$$\min_{\beta} \sum_{i=1}^{h} (r)^2{}_i$$

where $r^2{}_i \leq \cdots \leq r^2{}_n$ are the ordered squared residuals. This “trimming” approach effectively ignores extreme values, allowing for a breakdown point of up to 50% (Rousseeuw, 1984).

- Theil’s Pair-wise Median: This non-parametric estimator calculates the median of all possible slopes between pairs of observations:

$$\hat{\beta} = median\left\{\frac{y_j - y_i}{x_j - x_i} : x_j \neq x_i; 1 \leq i \leq j \leq n.\right\}$$

Theil’s method is rank-invariant and provides high robustness against contamination, regardless of the distribution of the error terms (Theil, 1950).

- Bayesian Inference (BAYES): We utilize a Bayesian framework with a Normal likelihood $Y \sim N(X\beta, \sigma^2 I)$. The posterior is proportional to the likelihood multiplied by the prior $p(\beta|y) \,\alpha\, p(y|\beta)$ x $p(\beta)$. We employ Markov Chain Monte Carlo (MCMC) sampling to approximate the posterior. It is noted that with non-informative priors, this model mirrors the vulnerability of OLS to outliers, serving as a control to test the framework’s inherent behaviours in the absence of robust priors.

3.3 Performance Measures

To evaluate the estimators, we compute the Bias, Mean Square Error (MSE), and Monte Carlo Standard Error (MCSE). Estimators are ranked using the Relative Mean Square Error (RelMSE), defined as:

$$RelMSE = \frac{MSE_{ESTIMATOR}}{MSE_{OLSE}}$$

An RMSE < 1 indicates that the robust estimator outperforms the OLS baseline, while an RMSE > 1 indicates poorer performance relative to OLS.

Table 2: Estimators Time Complexity

Operating System	Windows 10	**n**	**Time (s)**			
GPU	None		**OLS**	**LTS**	**Theil**	**Bayesian**
Processor	AMD64	**10**	0.7650452	0.78072263	3.00598547	0.4356163
Physical Cores	4	**30**	0.8330894	2.4126984	27.1326169	0.6341675
Total Cores (Logical)	4	**50**	0.7973745	2.0737926	137.186096	0.81378903
CPU Frequency (Mhz)	1800	**100**	1.23786047	3.98561647	209.376108	1.4414487
Total RAM Available (GB)	10.90	**500**	7.6793758	23.3622947	3256.40775	6.44510657
Available RAM (GB)	4.99	**1000**	14.6509314	48.5905287	13514.4525	25.5686688

4.0 SIMULATION RESULTS AND DISCUSSIONS

4.1 Comparative Analysis of Error Distribution Empirical Behaviour

The standard model serves as the experimental control, reflecting the uncontaminated behaviour of the three underlying error distributions as sample size increases. The contaminations model introduces subtle but structured anomalies into the simulated error distributions, thereby testing the immediate resilience of the distributions. The Outliers Model evaluates the impact of isolated, high-leverage perturbations injected into the simulated datasets. The Mixtures model represents the most severe form of disruption,

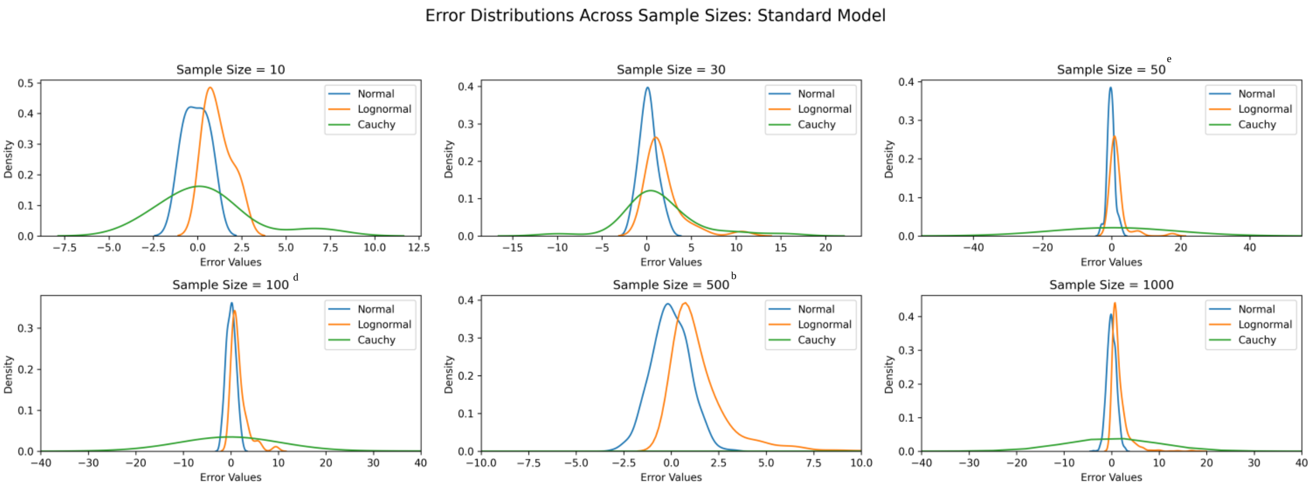

Figure 1: KDE Plot of Simulated Error for the Standard Error Model

concentrated near the origin, consistent with the Central Limit Theorem. The Lognormal distribution, while sharing a similar central tendency with the Normal in these models exhibits early signs of vulnerability.

At $n = 10$ and $n = 50$, the positive tail displays structural irregularities, including a distinct secondary shoulder and a prolonged extension into the upper support region. The distinct right-skewness at lower sample sizes ($n \leq 50$) characterized by a sharp positive gradient and a prolonged, asymmetric right tail signifies that contamination amplifies the

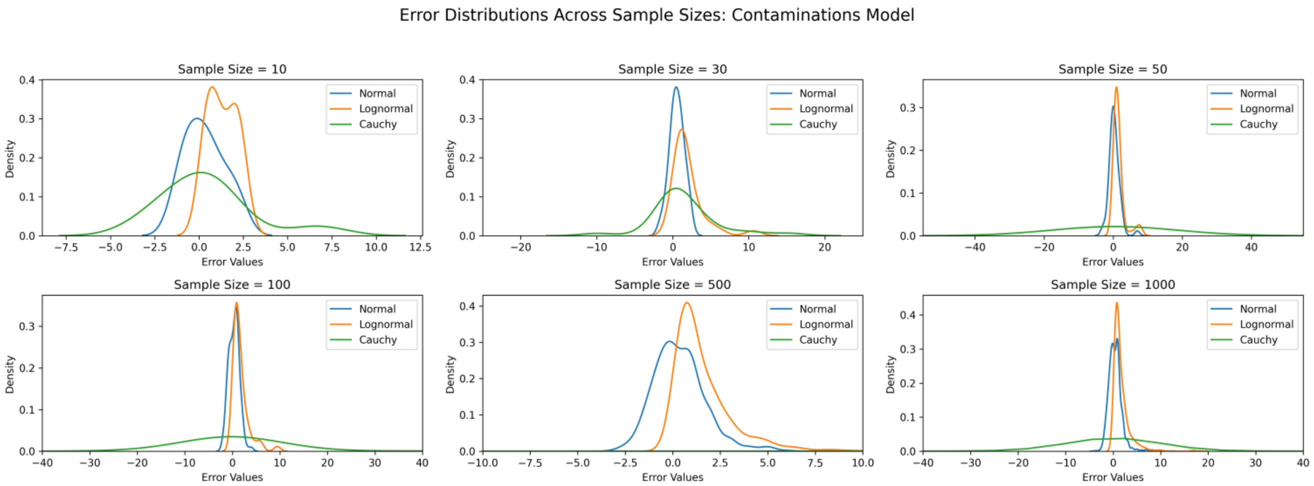

Figure 2: KDE Plot of Simulated Error for the Contaminations Error Model

Even at $n = 10$, the Cauchy distribution exhibits a suppressed peak density (approximately 0.15) and a broad support base relative to the Gaussian baseline. As the sample size increases to $n = 50, 100, 500,$ and 1000, the coordinate space stabilizes with the interval $[-40, 40]$, while the density profile remains flat, highly dispersed, and constrained to a peak density below 0.05, thereby confirming the infinite variance property characteristic of the Cauchy distribution. At larger sample sizes ($n \geq 500$), the Cauchy distribution dominates the geometry of the error space of the contaminations model. The extreme flatness of the density curve across the x-axis scale indicates that contamination further intensifies tail-weighting effect, causing the comparatively cleaner Gaussian and Lognormal distributions to collapse visually into a single, co-located spike around the origin.

The Outliers model highlights the stark contrast between light-tailed (Normal) and heavy-tailed (Cauchy) distributions. In these plots, the Cauchy distribution serves as the reference for robustness. While the Normal distribution attempts to force convergence around a zero-mean spike, it remains highly sensitive to the extreme values inherent in the Outlier model. At small sample sizes ($n = 10$), the Normal distribution develops a distinct secondary mode in the positive domain near the value of 10. This localized peak provides a direct visualization of the outlier point-mass, which has not yet been averaged out due to the small sample size.

As the sample size increases ($n = 30, 50, 100$), this secondary mode gradually transforms into a sequence of tail ripples along the positive tail of the Normal distribution. By $n = 500$ and $n = 1000$, these outlier-induced peaks becomes visually absorbed into the primary central mode for the Gaussian distribution, although they continue to leave behind an inflated tail structure.

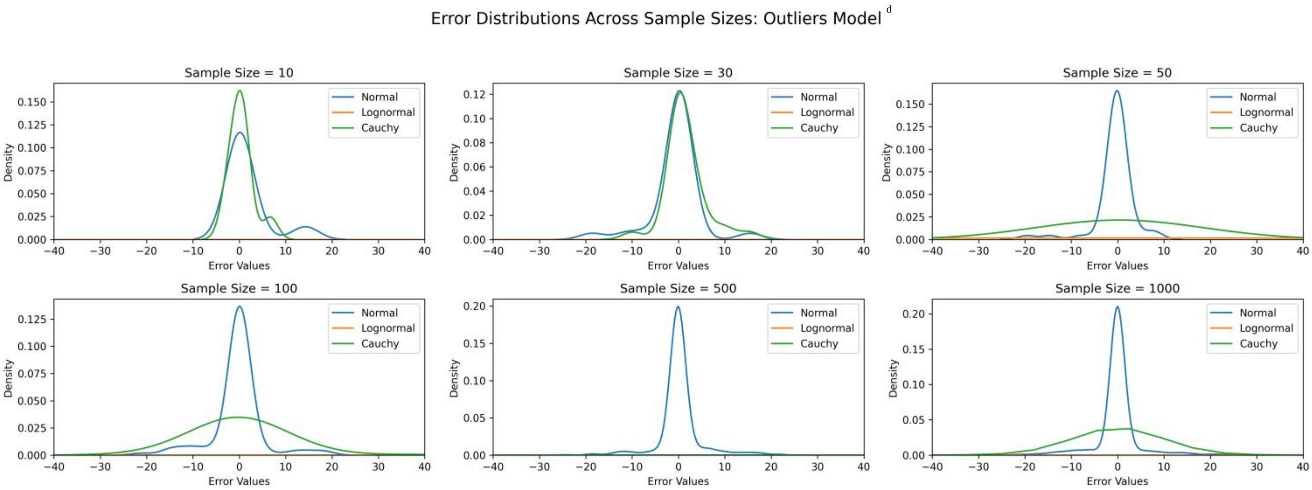

Figure 3: KDE Plot of Simulated Error for the Outliers Error Model

Conversely, the Cauchy distribution demonstrates significant dispersion across all sample sizes. Its intrinsic tail thickness is already so substantial that the introduction of point-mass outliers fails to produce any noticeable alteration in its empirical density profile. Even at $n = 1000$, the Cauchy KDE shows a broad, flat distribution (platykurtic) compared to the sharp spike of the Normal distribution. This confirms that for data generating processes susceptible to extreme outliers, classical estimators based on Normal assumptions (like the mean) will likely suffer from high variance, whereas heavy-tailed estimators would be required to maintain stability.

The Mixtures model presents the most significant deviation from expected asymptotic behaviour. At low sample sizes ($n = 10, 30, 50$), the distributions for all three types are incredibly flat, indicating high epistemic uncertainty and a failure of the estimators to identify a singular central tendency. At $n \geq 50$, the Normal error distribution exhibits clear evidence of mixture behaviour, with well-defined secondary modes emerging between the values of ± 15 and ± 35. This confirms that the error-generating mechanism is sampling from distinct, non-homogeneous states. For larger samples ($n \geq 100$), the scale returns to the standard $[-40, 40]$. However, the Normal distribution continues to exhibit persistent tail volatility, characterized by asymmetrical ripples and minor modes in the positive tail.

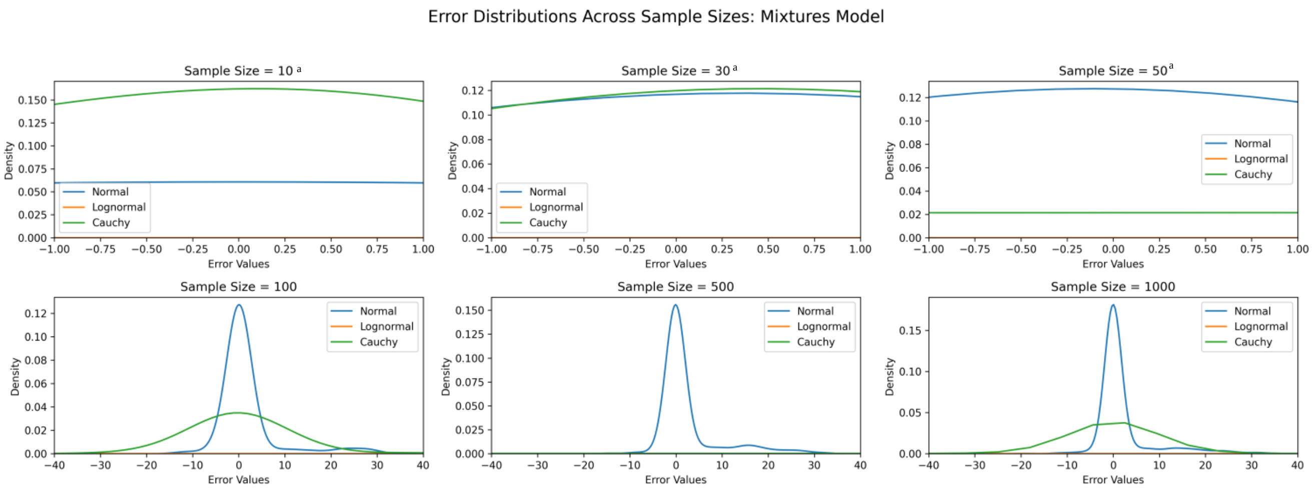

Figure 4: KDE Plot of Simulated Error for the Mixtures Error Model

X-axis scale of the figure is truncated at: [-1, 1][a], [-10, 10][b], [-25, 25][c], [-40, 40][d] and [-55, 55][e]

Unlike the other models, the Mixture plots show a resolution phase. It is not until $n \geq 100$ that the distributions begin to tighten. Specifically, the Normal distribution exhibits a distinct, sharp peak, yet the residual density in the tails (especially for the Cauchy and Lognormal profiles) suggests the presence of multi-modal components. This implies that for mixture-based data, small-sample inference is unreliable, and a much larger sample size is required to achieve stable parameter estimation compared to the Standard or Contamination models.

The Lognormal distribution, under both the Outliers and Mixtures models, becomes essentially flattened (approaching a density of zero) indicating a severe case of variance inflation and numerical divergence. The Outliers and Mixtures models are designed to generate data with specific local structures - clusters, heavy tails, or specific mixture weights. The fact that the Lognormal distribution cannot fit these structures, while the Cauchy (heavy-tailed) and Normal (peaked) distributions maintain at least some visible density, highlights that the Lognormal distribution is mathematically rigid in its inability to accommodate the specific types of residuals produced by these models.

From a researcher's perspective, the flattening of the Lognormal distribution is a statistical indicator that the Lognormal distribution should be discarded as a candidate model for these specific data-generating processes. It confirms that for data characterized by either extreme outliers or complex mixture components, the Lognormal model fails to provide a viable representation of the probability space. In such studies, researchers should prioritize distributions that can either handle heavy tails (like the Cauchy or Student-t distributions) or model multi-modal clusters (like Gaussian Mixture Models), rather than attempting to force a fit using a Lognormal assumption.

The severe scale inflation observed in the mixtures model, together with the persistent dispersion of the Cauchy distribution across all simulation frameworks, highlights the risks associated with relying on standard Ordinary Least Squares (OLS) estimation. These behaviours confirm the breakdown of the assumptions of finite variance and normality. Consequently, the visual evidence provided by these KDE plots leads to three primary conclusions regarding model robustness:

Table 3a: General Estimators' Performance Results Chart for Some Regression Estimators (Sample Size n=10)

β								
n = 10	**STANDARD MODEL**							
Method	**b**	**MCSE**	**Bias**	**MSE**	**RelMSE**	**LCL**	**UCL**	**Cov. Pr.**
Normal								
OLS	2.000	0.000	0.000	0.012	1.000	1.921	2.079	1.000
Bayesian	1.976	0.000	-0.024	0.012	1.024	1.898	2.054	1.000
Theil	2.000	0.000	0.000	0.014	1.137	1.916	2.084	1.000
LTS	2.000	0.000	0.000	0.017	1.418	1.906	2.094	1.000
Lognormal								
Theil	2.000	0.000	0.000	0.016	0.286	1.909	2.090	1.000
LTS	2.000	0.000	0.000	0.026	0.464	1.885	2.115	1.000
Bayesian	1.975	0.001	-0.025	0.055	0.987	1.808	2.143	1.000
OLS	1.999	0.001	-0.001	0.056	1.000	1.830	2.168	1.000
Cauchy								
Theil	2.000	0.001	0.000	0.097	0.000	1.777	2.223	1.000
LTS	1.997	0.002	-0.003	0.351	0.000	1.573	2.421	1.000
Bayesian	1.734	1.189	-0.266	2.1E+05	0.976	-327.693	331.162	1.000
OLS	1.755	1.203	-0.245	2.2E+05	1.000	-331.665	335.176	1.000
OUTLIERS MODEL								
Method	**b**	**MCSE**	**Bias**	**MSE**	**RelMSE**	**LCL**	**UCL**	**Cov. Pr.**
Normal								
Theil	2.000	0.001	0.000	0.081	0.168	1.796	2.203	1.000
LTS	2.000	0.001	0.000	0.163	0.338	1.711	2.289	1.000
Bayesian	1.975	0.002	-0.025	0.472	0.977	1.484	2.466	1.000
OLS	1.999	0.002	-0.001	0.483	1.000	1.502	2.496	1.000
Lognormal								
Theil	2.090	0.001	0.090	0.118	0.000	1.853	2.327	1.000
LTS	2.1E+06	1.5E+06	2.1E+06	3.3E+17	0.000	-4.1E+08	4.1E+08	1.000
Bayesian	8.6E+13	5.8E+13	8.6E+13	5.0E+32	0.976	-1.6E+16	1.6E+16	1.000
OLS	8.7E+13	5.9E+13	8.7E+13	5.1E+32	1.000	-1.6E+16	1.6E+16	1.000
Cauchy								
Theil	2.000	0.001	0.000	0.097	0.000	1.777	2.223	1.000
LTS	1.997	0.002	-0.003	0.351	0.000	1.573	2.421	1.000
Bayesian	1.734	1.189	-0.266	2.1E+05	0.976	-327.693	331.162	1.000
OLS	1.755	1.203	-0.245	2.2E+05	1.000	-331.665	335.176	1.000

α								
n = 10	**STANDARD MODEL**							
Method	**a**	**MCSE**	**Bias**	**MSE**	**RelMSE**	**LCL**	**UCL**	**Cov. Pr.**
Normal								
OLS	5.000	0.002	0.000	0.469	1.000	4.510	5.490	1.000
LTS	5.002	0.002	0.002	0.661	1.400	4.420	5.584	1.000
Theil	4.998	0.002	-0.002	0.868	1.900	4.332	5.665	0.572
Bayesian	3.677	0.002	-1.323	2.193	4.700	3.201	4.153	1.000
Lognormal								
Bayesian	5.178	0.004	0.178	2.085	0.400	4.153	6.203	0.997
LTS	6.204	0.003	1.204	2.462	0.500	5.484	6.924	0.999
Theil	6.613	0.003	1.613	4.166	0.900	5.718	7.508	0.148
OLS	6.650	0.004	1.650	4.899	1.000	5.595	7.705	0.992
Cauchy								
Theil	5.002	0.006	0.002	5.944	0.000	3.258	6.746	0.684
LTS	5.024	0.010	0.024	14.720	0.000	2.280	7.769	0.989
OLS	6.906	2.500	1.906	9.4E+05	1.000	-685.767	699.579	0.881
Bayesian	5.516	2.714	0.516	1.1E+06	1.200	-746.320	757.353	0.894
OUTLIERS MODEL								
Method	**a**	**MCSE**	**Bias**	**MSE**	**RelMSE**	**LCL**	**UCL**	**Cov. Pr.**
Normal								
LTS	5.000	0.004	0.000	2.117	0.300	3.960	6.041	0.985
Theil	4.285	0.004	-0.715	2.843	0.500	3.192	5.377	0.724
OLS	5.004	0.006	0.004	6.159	1.000	3.229	6.780	0.942
Bayesian	3.681	0.007	-1.319	8.261	1.300	1.855	5.508	0.971
Lognormal								
Theil	6.115	0.005	1.115	5.139	0.000	4.703	7.527	0.626
LTS	-7.0E+06	4.9E+06	-7.0E+06	3.6E+18	0.000	-1.4E+09	1.4E+09	0.990
OLS	-3.1E+14	2.1E+14	-3.1E+14	6.9E+33	1.000	-6.0E+16	5.9E+16	0.491
Bayesian	-3.2E+14	2.2E+14	-3.2E+14	7.3E+33	1.100	-6.1E+16	6.1E+16	0.480
Cauchy								
Theil	5.002	0.006	0.002	5.944	0.000	3.258	6.746	0.684
LTS	5.024	0.010	0.024	14.720	0.000	2.280	7.769	0.989
OLS	6.906	2.500	1.906	9.38E+05	1.000	-685.767	699.579	0.881
Bayesian	5.516	2.714	0.516	1.10E+06	1.200	-746.320	757.353	0.894

- The rapid tightening of the Normal distribution across all models suggests that if the underlying process is even remotely Gaussian, standard estimators will converge quickly. However, this convergence is deceptive in the presence of outliers or mixture components, where the peak may not represent the true global parameter.
- The Cauchy distribution's persistent dispersion across all sample sizes indicates that it remains an effective, though broad, characterization of error when the data contains extreme values. Researchers should prefer Cauchy-based or robust regression techniques when the visual profile of the data (or diagnostic residual plots) mirrors the broad, flat structure observed in the Outliers or Mixtures models.
- There is a discernible critical threshold for sample sizes. In Standard and Contamination models, $n = 50$ appears sufficient for reliable density estimation. However, in Mixtures and Outlier-heavy models, meaningful convergence is not observed until $n \geq 100$. This suggests that for complex data structures, increasing sample size is not merely a method to reduce variance but a prerequisite to resolving the underlying distribution's true shape.

4.2.0 PARAMETER ESTIMATION

4.2.1 Estimators' Performance

The performance of the Slope β) and Intercept (α) estimators - Ordinary Least Squares (OLS), Least Trimmed Squares (LTS), Theil's estimator, and Bayesian estimation - was evaluated across standard and contaminated distributions using metrics including Bias, Mean Square Error (MSE), Root Mean Square Error (RMSE), Monte Carlo Standard Error (MCSE), and Coverage Probability (Cov. Pr.). Under standard Normal conditions, OLS consistently demonstrated optimal performance, exhibiting minimal bias and negligible MSE across all tested sample sizes. While robust estimators and Bayesian methods performed comparably to OLS in terms of bias, small variations in RMSE were observed, particularly at smaller sample sizes of $n = 10$ and $n = 30$. Bias values for slope estimators were generally

Table 3b: Table General Estimators' Performance Results Chart for Some Regression Estimators (Sample Size n=10)

β								
n = 10	**MIXTURES MODEL**							
Method	**β**	**MCSE**	**Bias**	**MSE**	**RelMSE**	**LCL**	**UCL**	**Cov. Pr.**
				Normal				
Theil	2.257	0.001	0.257	0.137	0.096	2.066	2.448	1.000
LTS	2.520	0.001	0.520	0.581	0.409	2.121	2.919	1.000
Bayesian	2.933	0.002	0.933	1.342	0.944	2.442	3.424	1.000
OLS	2.969	0.002	0.969	1.421	1.000	2.472	3.466	1.000
				Lognormal				
Theil	2.364	0.001	0.364	0.350	0.000	2.030	2.698	1.000
LTS	4.7E+10	3.3E+10	4.7E+10	1.6E+26	0.00	-9.0E+12	9.1E+12	1.000
Bayesian	1.9E+18	1.3E+18	1.9E+18	2.4E+41	0.98	-3.5E+20	3.6E+20	1.000
OLS	1.9E+18	1.3E+18	1.9E+18	2.5E+41	1.00	-3.6E+20	3.6E+20	1.000
				Cauchy				
Theil	2.000	0.001	0.000	0.097	0.000	1.777	2.223	1.000
LTS	1.997	0.002	-0.003	0.351	0.000	1.573	2.421	1.000
Bayesian	1.734	1.189	-0.266	2.1E+05	0.976	-327.693	331.162	1.000
OLS	1.755	1.203	-0.245	2.2E+05	1.000	-331.665	335.176	1.000
				CONTAMINATIONS MODEL				
Method	**β**	**MCSE**	**Bias**	**MSE**	**RelMSE**	**LCL**	**UCL**	**Cov. Pr.**
				Normal				
Theil	2.169	0.000	0.169	0.043	0.791	2.083	2.255	1.000
LTS	2.168	0.000	0.168	0.044	0.812	2.078	2.259	1.000
Bayesian	2.168	0.000	0.168	0.045	0.818	2.075	2.260	1.000
OLS	2.194	0.000	0.194	0.055	1.000	2.101	2.287	1.000
				Lognormal				
Theil	2.047	0.000	0.047	0.016	0.358	1.962	2.131	1.000
LTS	2.051	0.000	0.051	0.025	0.559	1.943	2.158	1.000
Bayesian	2.009	0.001	0.009	0.043	0.953	1.861	2.158	1.000
OLS	2.034	0.001	0.034	0.045	1.000	1.884	2.184	1.000
				Cauchy				
Theil	2.000	0.001	0.000	0.097	0.000	1.777	2.223	1.000
LTS	1.997	0.002	-0.003	0.351	0.000	1.573	2.421	1.000
Bayesian	1.734	1.189	-0.266	2.1E+05	0.976	-327.693	331.162	1.000
OLS	1.755	1.203	-0.245	2.2E+05	1.000	-331.665	335.176	1.000

α								
n = 10	**MIXTURES MODEL**							
Method	**α**	**MCSE**	**Bias**	**MSE**	**RelMSE**	**LCL**	**UCL**	**Cov. Pr.**
				Normal				
Theil	3.464	0.004	-1.536	4.535	0.300	2.409	4.520	0.884
LTS	3.311	0.005	-1.689	6.555	0.400	1.935	4.687	1.000
OLS	1.671	0.006	-3.329	17.243	1.000	-0.105	3.446	0.998
Bayesian	0.230	0.007	-4.770	29.274	1.700	-1.597	2.057	0.999
				Lognormal				
Theil	4.611	0.006	-0.389	6.430	0.000	2.819	6.404	0.943
LTS	-1.5E+11	1.1E+11	-1.5E+11	1.8E+27	0.000	-3.0E+13	3.0E+13	0.970
OLS	-6.9E+18	4.7E+18	-6.9E+18	3.3E+42	1.00	-1.3E+21	1.3E+21	0.099
Bayesian	-7.1E+18	4.9E+18	-7.1E+18	3.5E+42	1.10	-1.4E+21	1.3E+21	0.093
				Cauchy				
Theil	5.002	0.006	0.002	5.944	0.000	3.258	6.746	0.684
LTS	5.024	0.010	0.024	14.720	0.000	2.280	7.769	0.989
OLS	6.906	2.500	1.906	9.38E+05	1.000	-685.767	699.579	0.881
Bayesian	5.516	2.714	0.516	1.10E+06	1.200	-746.320	757.353	0.894
				CONTAMINATIONS MODEL				
Method	**α**	**MCSE**	**Bias**	**MSE**	**RelMSE**	**LCL**	**UCL**	**Cov. Pr.**
				Normal				
OLS	4.334	0.002	-0.666	0.970	1.000	3.815	4.852	1.000
LTS	4.430	0.002	-0.570	0.980	1.010	3.851	5.008	1.000
Theil	4.067	0.002	-0.933	1.723	1.777	3.406	4.727	0.916
Bayesian	2.987	0.002	-2.013	4.555	4.696	2.480	3.495	1.000
				Lognormal				
Bayesian	5.055	0.004	0.055	1.900	0.434	4.070	6.040	0.997
LTS	6.019	0.002	1.019	1.970	0.450	5.329	6.709	0.999
Theil	6.355	0.003	1.355	3.320	0.759	5.483	7.226	0.211
OLS	6.532	0.004	1.532	4.376	1.000	5.513	7.551	0.993
				Cauchy				
Theil	5.002	0.006	0.002	5.944	0.000	3.258	6.746	0.684
LTS	5.024	0.010	0.024	14.720	0.000	2.280	7.769	0.989
OLS	6.906	2.500	1.906	9.4E+05	1.000	-685.767	699.579	0.881
Bayesian	5.516	2.714	0.516	1.1E+06	1.178	-746.320	757.353	0.894

centred near zero. Furthermore, coverage probabilities for the Normal distribution were consistently high, often approaching $n = 100$, which indicates that the constructed confidence intervals effectively captured the true parameters. As sample sizes increased to $n = 1000$ interval widths tightened, reflecting increased precision in parameter estimation.

4.2.2 Effects of Contaminations on Estimators' Performance

The introduction of non-normal distributions, specifically the Lognormal model, significantly altered the performance metrics of the estimators compared to the standard Normal model. Under Lognormal distributions, the Intercept (α) estimators experienced a notable increase in Bias and MSE across almost all methods. In the presence of such contamination, OLS often exhibited higher susceptibility to outliers, particularly regarding the intercept, where Bias and MSE were markedly elevated compared to the results under Normal conditions. In contrast, LTS and Theil's estimators generally provided more stable results when faced with Lognormal contamination. While bias remained present for all estimators, these robust methods often maintained lower MSE values in slope parameter estimation compared to the standard OLS approach, demonstrating superior resilience to the skewed nature of the Lognormal distribution.

4.2.3 Effects of Sample Size on Estimators' Performance

The simulation results provide clear evidence of the asymptotic properties of the estimators as sample sizes increased from $n = 10$ to $n = 1000$. For all methods, there was a pronounced reduction in both MSE and MCSE as the sample size increased, which confirms the consistency of the estimators. At small sample sizes ($n \leq 50$), estimators exhibited higher volatility, particularly in the intercept, which showed higher bias that was often mitigated as n grew. Differences in performance between the estimators were more pronounced at these levels, with robust estimators providing better protection against potential outliers compared to OLS. At larger sample sizes of $n = 500$ and $n = 1000$, the estimators reached a high degree of stability, with consistently lower MCSE values and robust coverage probabilities. The performance gap between robust methods and OLS became more distinct as sample size allowed for better identification of the underlying distribution, highlighting the consistency and reliability of the methods employed.

4.2.4. Implications for Research Methodology

For empirical research involving these distributions, the findings suggest that:

- When residual distribution is unknown, Theil's estimator is the most reliable default choice. Its performance consistently sits in the upper tier across all models, providing the best protection against unknown contamination without the high-variance penalty of purely parametric robust models.
- When the assumption of normality is confirmed, OLS remains the preferred estimator. The efficiency gap between OLS and robust estimators is non-trivial; using a robust estimator like LTS on clean data results in quantifiable losses in precision (wider confidence intervals).
- For complex mixture processes**,** neither the simple frequentist estimators nor standard Bayesian models are sufficient. The plateauing of error metrics suggests that the complexity of the *Mixtures* model requires a transition from simple estimation to robust estimation where the estimator itself is designed to capture the multi-modality of the underlying process.

4.3 Empirical Data Analysis

Gross Domestic Products (Current GDP US$) and total Population for 266 countries for the year 2018 and 2024 were retrieved from the World Development Indicators (WDI) data website (World Bank, 2026). Figure 5 presents the scatter plot of the dataset before and after contaminating 20% of the observations respectively. Figure 6 and 7 show that the OLS assumptions of normally distributed residuals and homogeneous variance (homoscedasticity) are violated.

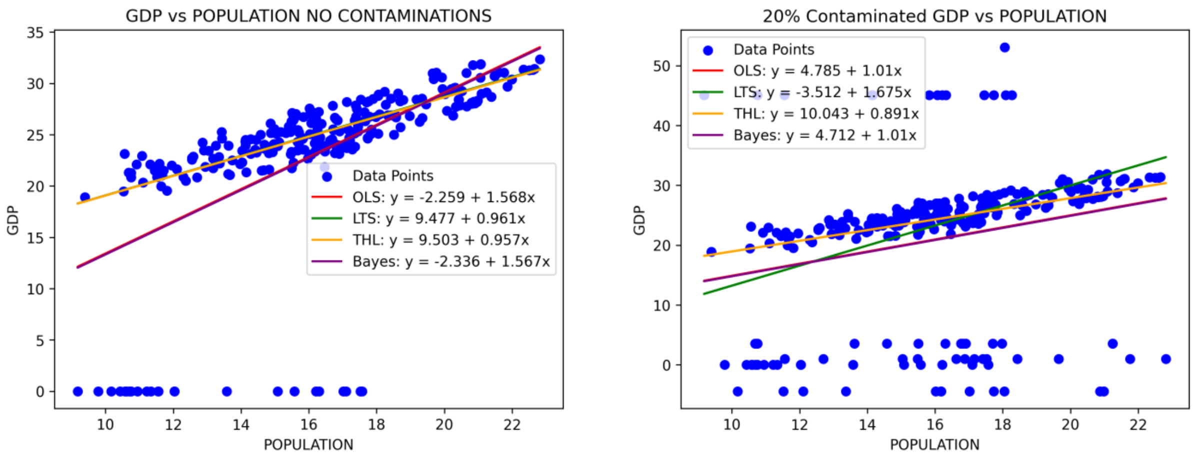

Figure 5: Scatter Plot of GDP (US$) vs Total Population

Regression analysis was conducted on the dataset for $n = 10$ countries, $n = 50$ countries and $n = 266$ countries respectively. Table 6 presents the regression parameter estimates obtained using the previously discussed estimation methods. The prior parameters for the intercept ($\alpha = 8.400$) and slope parameters ($\beta = 0.997$) were obtained from the OLS estimates of the dataset for the year 2018.

The efficacy of the estimators exhibits significant dependence upon both sample size (n) and the presence of data contamination. In uncontaminated datasets, both the Ordinary Least Squares Estimator (OLS) and the Bayesian estimator demonstrate high precision, characterized by lower bias relative to robust alternatives. Furthermore, these estimators produce tighter credible intervals in the absence of outliers, underscoring their statistical efficiency as unbiased estimators for normal distributions.

The Mean Square Error (MSE), mathematically defined as the sum of estimator variance and the square of the bias, (MSE = Variance + Bias^2), serves as a primary metric for evaluating total estimator error. In clean data environments ($n = 266$), OLS and Bayesian estimators yield MSE values of approximately 40.277 and 40.283, respectively, outperforming Theil and Least Trimmed Squares (LTS), which report higher MSE values of approximately 50.816 and 51.016. This performance gap illustrates the deliberate trade-off inherent in robust estimation: methods like LTS and Theil intentionally sacrifice a degree of efficiency (manifested as higher variance) in perfect data to secure substantial bias reduction in contaminated environments. With 20% contamination, the significant mitigation of bias facilitated by the LTS estimator outweighs the variance penalty, resulting in a more robust and reliable fit than that provided by the compromised OLS or Bayesian models.

Table 6a: Analysis Results for Empirical Data with no Contaminations

	α			C. Int. (α)		β			C. Int. (β)		MODEL	
	$\bar{x}$	BIAS	σ	LCL	UCL	$\bar{x}$	BIAS	σ	LCL	UCL	BIAS	MSE
	DATA WITH NO CONTAMINATIONS											
ESTIMATORS	n=10											
OLS	-0.494	8.893	15.003	-2.271	1.284	1.291	-0.293	0.907	-0.487	3.068	0.000	108.268
Bayes	-1.544	9.943	15.105	-3.333	0.246	1.289	-0.291	0.913	-0.501	3.078	1.086	110.921
Theil	11.177	-2.777	16.196	9.258	13.095	0.771	0.226	0.979	-1.147	2.690	-3.282	136.949
LTS	13.761	-5.361	17.185	11.725	15.796	0.701	0.296	1.039	-1.334	2.737	-4.738	164.505
	n=50											
OLS	-5.435	13.835	5.779	-6.151	-4.720	1.759	-0.762	0.365	1.044	2.475	0.000	49.663
Bayes	-5.670	14.070	5.783	-6.386	-4.954	1.757	-0.760	0.365	1.041	2.473	0.267	49.808
Theil	10.175	-1.775	6.323	9.392	10.958	0.880	0.117	0.399	0.098	1.663	-1.910	63.102
LTS	11.436	-3.036	6.508	10.630	12.241	0.841	0.157	0.411	0.035	1.646	-2.549	69.477
	n=266											
OLS	-2.260	10.659	2.088	-2.507	-2.012	1.568	-0.570	0.127	1.320	1.816	0.000	40.277
Bayes	-2.311	10.711	2.088	-2.559	-2.063	1.567	-0.570	0.127	1.320	1.815	0.056	40.283
Theil	9.503	-1.103	2.263	9.234	9.772	0.957	0.040	0.137	0.688	1.226	-1.863	50.816
LTS	9.477	-1.077	2.265	9.208	9.746	0.961	0.037	0.137	0.692	1.230	-1.896	51.016

The introduction of 20% contamination elucidates the classic bias-variance trade-off within statistical modelling. While OLS remains the most efficient estimator under Gaussian assumptions, its predictive accuracy degrades precipitously in the presence of outliers. For instance, in a contaminated sample size of $n = 10$, the OLS estimator exhibits a marked escalation in bias compared to its performance within a clean data context. Visual inspection of the scatter plots in figure 5 above further corroborates these findings. While the OLS regression lines align closely with clean data, they are significantly displaced by outliers in the contaminated dataset; conversely, robust estimators such as Theil and LTS consistently maintain a trend trajectory proximate to the central mass of the data.

The deleterious effects of 20% contamination are starkly evidenced by comparative analysis of the residuals and Normal P-P plots. In contaminated scenarios, OLS and Bayesian estimators exhibit a pronounced divergence from the reference line (y=x), underscoring their limited capacity to accommodate heavy-tailed or outlier-laden distributions. In contrast, robust estimators demonstrate greater fidelity to the underlying data trend.

Table 6b: Analysis Results for Empirical Data With 20% Contaminations

	α			**C. Int. (α)**		**β**			**C. Int. (β)**		**MODEL**	
	$\bar{x}$	**BIAS**	**σ**	**LCL**	**UCL**	**$\bar{x}$**	**BIAS**	**σ**	**LCL**	**UCL**	**BIAS**	**MSE**
	DATA WITH 20% CONTAMINATIONS											
ESTIMATORS	n=10											
OLS	17.022	-8.622	26.529	13.922	20.122	0.239	0.759	1.582	-2.861	3.339	0.000	138.774
Bayes	15.654	-7.255	26.685	12.536	18.773	0.252	0.745	1.591	-2.866	3.371	1.145	141.723
Theil	14.593	-6.193	28.453	11.268	17.918	0.625	0.372	1.696	-2.700	3.950	-3.982	175.492
LTS	12.191	-3.791	29.868	8.701	15.681	0.847	0.151	1.781	-2.643	4.337	-5.259	203.560
	n=50											
OLS	10.495	-2.095	9.068	9.434	11.556	0.722	0.276	0.541	-0.339	1.783	0.000	126.949
Bayes	10.210	-1.810	9.071	9.149	11.271	0.722	0.275	0.542	-0.339	1.784	0.275	127.103
Theil	11.965	-3.565	9.473	10.856	13.073	0.834	0.164	0.566	-0.275	1.942	-3.318	149.541
LTS	11.656	-3.256	9.520	10.542	12.769	0.864	0.134	0.568	-0.250	1.978	-3.502	152.170
	n=266											
OLS	4.785	3.615	3.911	4.320	5.249	1.010	-0.013	0.237	0.546	1.475	0.000	141.396
Bayes	4.737	3.663	3.912	4.272	5.201	1.010	-0.013	0.237	0.546	1.475	0.048	141.400
Theil	-3.512	11.912	4.054	-3.994	-3.031	1.675	-0.678	0.246	1.194	2.157	-2.488	158.057
LTS	10.043	-1.643	4.065	9.560	10.526	0.891	0.106	0.246	0.409	1.374	-3.327	163.749

The performance of the Least Trimmed Squares (LTS) estimator is particularly salient regarding sample size scalability. In small samples ($n = 10$), LTS displays high variability; the elevated bias observed in the intercept and slope relative to the non-contaminated baseline suggests that the trimming process may inadvertently discard critical information, thereby impeding reliable parameter estimation.

However, as the sample size increases to $n = 266$, the robustness of the LTS estimator becomes manifest. The estimator exhibits enhanced consistency and diminished sensitivity to 20% contamination compared to the $n = 10$ cohort. Consequently, with a larger dataset, the trimming mechanism operates more effectively to isolate the 80% of uncontaminated data, enabling the regression line to recover the true underlying trend despite the presence of outliers.

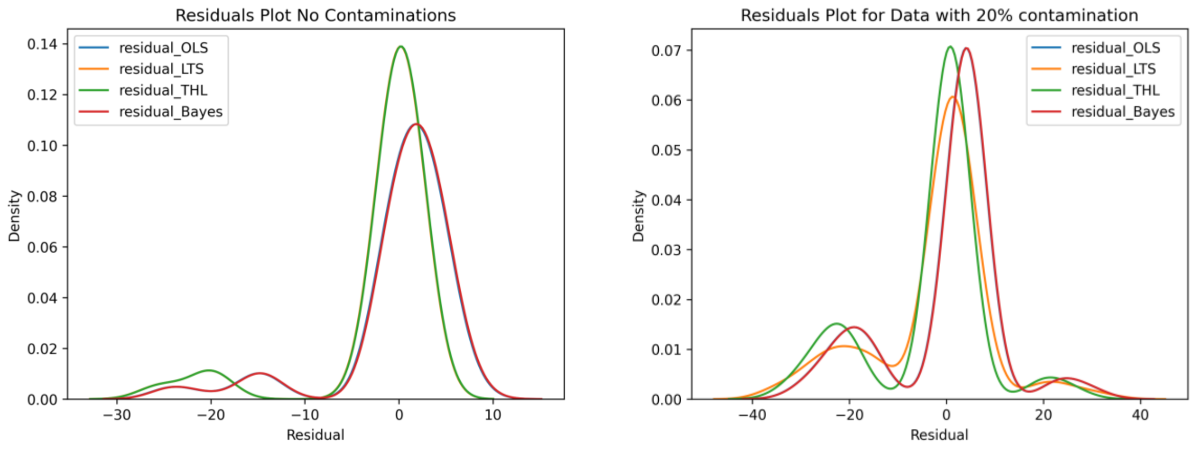

Figure 6: Residuals KDE Plot for estimators under study

4.4 Hypothesis Testing and Estimator Comparison

A two-sided hypothesis was set up for the regression slope parameter β as follows:

$$H_0 : \beta = 0 \tag{17}$$

Versus

$$H_1 : |\beta| > 0 \tag{18}$$

where $\beta \in \{1.289, 1.757, 1.567\}$, the OLS estimates for $n \in \{10, 50, 266\}$ respectively under no contaminations regime.

Looking at the range of the confidence intervals for the contaminated data presented in Table 6b above, constructed for all estimators under consideration at α=0.05, it is observed that, for small sample sizes ($n = 10$), LTS estimators demonstrated the highest precision relative to the null hypothesis by yielding an empirical mean slope (0.847) that is approximately closest to the hypothesized value ($\beta = 1.289$), followed closely by the Theil with an empirical mean slope of 0.625. The 95% confidence intervals for the Theil's [-2.700 – 3.950] and the LTS [-2.643 – 4.337] estimators both contained the hypothesized value. In contrast, the OLS and Bayesian estimators displayed substantial deviations from the hypothesized parameter in their point estimations, yielding an empirical mean slope of $\beta = 0.239$ and $\beta = 0.252$ respectively, though with a tighter 95% confidence intervals for the OLS [-2.861 – 3.339] and Bayesian [-2.866 – 3.371] estimator respectively.

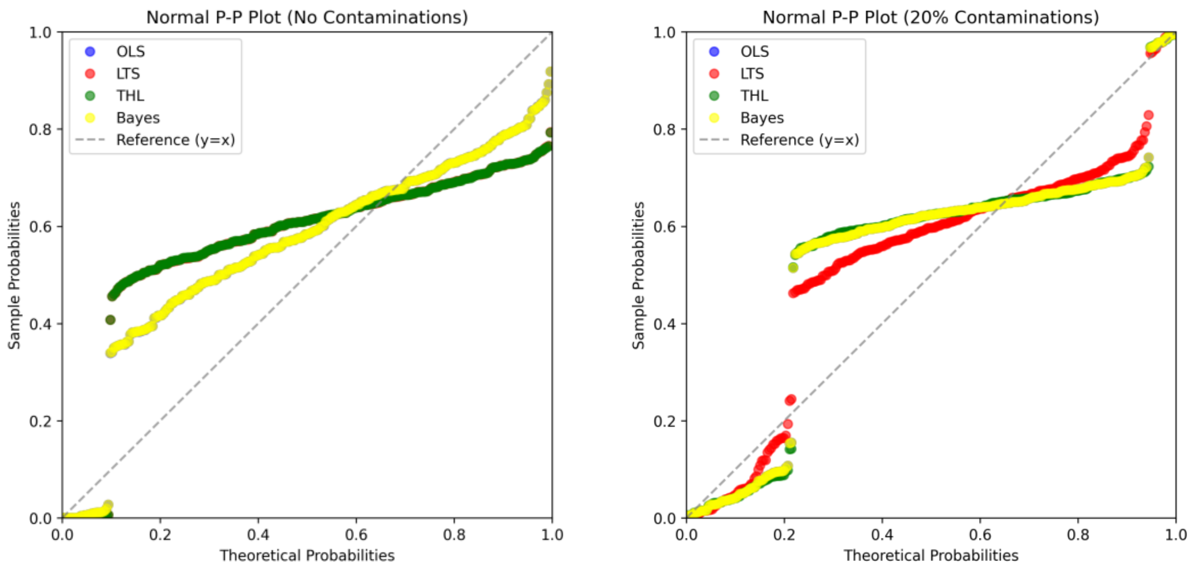

Figure 7: Normal P-P Plot for estimators under study

As the sample size increased ($n = 50$), all the estimators exhibited substantial improvements in precision. However, LTS estimator continued to demonstrate the highest precision, followed closely by the Theil estimator. The 95% confidence intervals for all estimators became significantly tighter and contained the hypothesized values of $\beta = 1.757$ for the sample size. The intervals also indicate that the value of β could potentially be zero and, in some cases, even negative.

There is, therefore, insufficient evidence to reject the null hypothesis H_0 ($\beta = 0$) for all the estimators. Consequently, the test is not statistically significant at the α=0.05 level for small sample cases ($n \leq 50$). Thus, we do not reject $H_0 : \beta = 0$ for all estimators and conclude that there is insufficient evidence to establish a predictive relationship between GDP and Population that is consistent with the ground truth at the α=0.05 level of significance in small sample conditions.

When $n = 266$, all confidence intervals became strictly positive with significantly narrower width, effectively excluding zero as a plausible value of β. However, only the confidence intervals for Theil [1.194 – 2.157] contained the hypothesized $\beta = 1.567$. The Theil method now yield the highest precision with an empirical mean slope ($\beta = 1.675$) compared to LTS ($\beta = 0.891$) with confidence interval [0.409 – 1.374]. OLS and Bayes converged to mean slope value ($\beta = 1.010$) with confidence interval [0.546 – 1.475].

There is, therefore, sufficient evidence to reject the null hypothesis H_0 ($\beta = 0$) for all estimators. Consequently, the test is statistically significant at the α=0.05 level for sample sizes $n > 50$. Hence, we fail to reject $H_1: |\beta| > 0$ and conclude that there is sufficient evidence to establish a significant predictive relationship between GDP and Population at the α=0.05 level of significance when the sample size is large.

Hence, the fitted Simple Linear Regression model for each of the estimation procedure is as follows:

OLS: $GDP = 4.785 + 1.010 POPULATION\ (TOTAL)$

LTS: $GDP = 4.737 + 0.891 POPULATION\ (TOTAL)$

Theil: $GDP = -3.512 + 1.675 POPULATION\ (TOTAL)$

Bayesian: $GDP = 10.043 + 1.010 POPULATION\ (TOTAL)$

5.0 CONCLUSION

The usual assumption in linear regression is that error terms follow a normal distribution, under which the OLS procedure performs efficiently. However, in practice, it is nearly impossible to obtain a dataset that fully satisfies the normality assumption. Since violations of this assumption often result in loss of efficiency for the OLS procedure, alternative regression techniques become necessary. In this study, three robust procedures for the simple linear regression model were investigated and compared with OLS under both Normal and non-Normal error conditions.

Results obtained from the slope estimators indicate that, across all cases considered, Theil's method demonstrated the strongest overall performance relative to OLS. Specifically, it consistently exhibited negligible bias and the smallest MSE values. The second-best performance was obtained from the Least Trimmed Squares (LTS) method.

The findings of this study further show that Bayesian linear regression provides a more optimal and safer alternative to OLS under normal error conditions, offering inference that is conditional on the observed data and exact without reliance on asymptotic approximations. Theil's estimator demonstrated high small-sample efficiency across most conditions, particularly when the variance of the error terms was not constant. Furthermore, Theil's estimation procedure consistently exhibited the strongest and most reliable performance compared to other estimators, and under the conditions, investigated in this study.

The empirical results also support the relative robustness of the Bayesian estimator to moderate contamination in the dataset. The Least Trimmed Squares method was found to be particularly effective when the error term is heavy-tailed and the sample size is sufficiently large. In contrast, the OLS procedure remained reliable only when the normality assumptions were reasonably satisfied.

The current study has also confirmed the applicability of Theil method under varying conditions and its robustness relative to several competing robust methods, particularly the LTS method, within the scope of the conditions investigated. However, for large samples $(n > 100)$, the Theil's estimator becomes computationally complex and expensive while the LTS gives a fair and efficient performance – complexity trade-off. More so, the Bayesian model adopted in this study assumes a normal likelihood and a conjugate normal prior, thereby restricting the Bayesian framework primarily to situations involving approximately normal data distributions.

In addition, the design variable X was assumed to be non-stochastic, while outliers were examined only in the vertical Y-direction. Further studies are therefore recommended for cases in which the design variable is stochastic, and where Bayesian models are constructed using non-normal priors and robust likelihood functions. Such models should also be comparatively evaluated against other robust estimation procedures, particularly Theil's method, under broader contamination, multiple regressors and non-normal conditions.

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7.0 APPENDIX

7.1 Simulation Results Table

β							
n = 100	**STANDARD MODEL**						
Method	**β**	**MCSE**	**Bias**	**MSE**	**RelMSE**	**LCL**	**UCL**	**Cov. Pr.**
				Normal				
OLS	2.000	0.000	0.000	0.000	1.000	1.999	2.001	1.000
Bayesian	2.000	0.000	0.000	0.000	1.000	1.999	2.001	1.000
Theil	2.000	0.000	0.000	0.000	1.058	1.999	2.001	1.000
LTS	2.000	0.000	0.000	0.000	1.359	1.999	2.001	1.000
				Lognormal				
Theil	2.000	0.000	0.000	0.000	0.151	1.999	2.001	1.000
LTS	2.000	0.000	0.000	0.000	0.542	1.999	2.001	1.000
OLS	2.000	0.000	0.000	0.000	1.000	1.999	2.001	1.000
Bayesian	2.000	0.000	0.000	0.000	1.000	1.998	2.001	1.000
				Cauchy				
LTS	2.000	0.000	0.000	0.000	0.000	1.996	2.004	1.000
Theil	2.000	0.000	0.000	0.000	0.000	1.999	2.001	1.000
Bayesian	1.940	0.026	-0.060	105.108	1.000	-0.094	3.974	1.000
OLS	1.940	0.026	-0.060	105.110	1.000	-0.094	3.974	1.000
				OUTLIERS MODEL				
Method	**β**	**MCSE**	**Bias**	**MSE**	**RelMSE**	**LCL**	**UCL**	**Cov. Pr.**
				Normal				
Theil	2.000	0.000	0.000	0.000	0.126	1.998	2.002	1.000
LTS	2.000	0.000	0.000	0.000	0.182	1.998	2.002	1.000
OLS	2.000	0.000	0.000	0.000	1.000	1.996	2.004	1.000
Bayesian	2.000	0.000	0.000	0.000	1.000	1.996	2.004	1.000
				Lognormal				
Theil	2.001	0.000	0.001	0.000	0.000	2.000	2.002	1.000
LTS	4.9E+07	3.9E+07	4.9E+07	2.3E+20	0.000	-3.0E+09	3.1E+09	1.000
OLS	3.5E+14	3.2E+14	3.5E+14	1.5E+34	1.000	-2.4E+16	2.5E+16	1.000
Bayesian	3.5E+14	3.2E+14	3.5E+14	1.5E+34	1.000	-2.4E+16	2.5E+16	1.000
				Cauchy				
LTS	2.000	0.000	0.000	0.000	0.000	1.996	2.004	1.000
Theil	2.000	0.000	0.000	0.000	0.000	1.999	2.001	1.000
Bayesian	1.940	0.026	-0.060	105.108	1.000	-0.094	3.974	1.000
OLS	1.940	0.026	-0.060	105.110	1.000	-0.094	3.974	1.000

α								
n = 100	**STANDARD MODEL**							
Method	**α**	**MCSE**	**Bias**	**MSE**	**RelMSE**	**LCL**	**UCL**	**Cov. Pr.**
				Normal				
OLS	5.001	0.001	0.001	0.041	1.000	4.961	5.041	1.000
LTS	5.000	0.001	0.000	0.055	1.400	4.954	5.047	1.000
Theil	4.998	0.002	-0.002	0.481	11.800	4.861	5.136	1.000
Bayesian	3.952	0.001	-1.048	1.138	28.000	3.912	3.992	1.000
				Lognormal				
Bayesian	5.585	0.001	0.585	0.530	0.200	5.499	5.671	0.950
LTS	6.135	0.001	1.135	1.384	0.500	6.073	6.196	0.865
OLS	6.649	0.001	1.649	2.910	1.000	6.563	6.736	0.247
Theil	6.646	0.003	1.646	3.751	1.300	6.444	6.849	1.000
				Cauchy				
LTS	4.996	0.003	-0.004	1.458	0.000	4.757	5.236	0.974
Theil	5.011	0.006	0.011	5.547	0.000	4.544	5.478	0.999
Bayesian	7.289	1.796	2.289	4.8E+05	1.000	-130.734	145.312	0.726
OLS	8.341	1.801	3.341	4.9E+05	1.000	-130.095	146.777	0.644
				OUTLIERS MODEL				
Method	**α**	**MCSE**	**Bias**	**MSE**	**RelMSE**	**LCL**	**UCL**	**Cov. Pr.**
				Normal				
LTS	5.000	0.001	0.000	0.111	0.200	4.934	5.067	1.000
OLS	5.001	0.002	0.001	0.469	1.000	4.865	5.137	0.961
Theil	4.998	0.002	-0.002	0.585	1.200	4.846	5.150	1.000
Bayesian	3.952	0.002	-1.048	1.571	3.400	3.816	4.089	0.999
				Lognormal				
Theil	6.600	0.003	1.600	3.635	0.000	6.395	6.806	1.000
LTS	-1.5E+09	1.2E+09	-1.5E+09	2.2E+23	0.000	-9.5E+10	9.2E+10	0.990
OLS	-1.1E+16	1.0E+16	-1.1E+16	1.6E+37	1.000	-8.1E+17	7.9E+17	0.002
Bayesian	-1.1E+16	1.0E+16	-1.1E+16	1.6E+37	1.000	-8.1E+17	7.9E+17	0.002
				Cauchy				
LTS	4.996	0.003	-0.004	1.458	0.000	4.757	5.236	0.974
Theil	5.011	0.006	0.011	5.547	0.000	4.544	5.478	0.999
Bayesian	7.289	1.796	2.289	4.8E+05	1.000	-130.734	145.312	0.726
OLS	8.341	1.801	3.341	4.9E+05	1.000	-130.095	146.777	0.644

Table 4a: General Estimators' Performance Results Chart for Some Regression Estimators (Sample Size n=100)

β								
n = 100	**MIXTURES MODEL**							
Method	**β**	**MCSE**	**Bias**	**MSE**	**RelMSE**	**LCL**	**UCL**	**Cov. Pr.**
Normal								
Theil	2.023	0.000	0.023	0.001	0.059	2.021	2.024	1.000
LTS	2.042	0.000	0.042	0.002	0.210	2.039	2.045	1.000
OLS	2.096	0.000	0.096	0.010	1.000	2.092	2.100	1.000
Bayesian	2.096	0.000	0.096	0.010	1.000	2.092	2.100	1.000
Lognormal								
Theil	2.023	0.000	0.023	0.001	0.000	2.021	2.025	1.000
LTS	1.1E+12	8.7E+11	1.1E+12	1.1E+29	0.000	-6.6E+13	6.8E+13	1.000
OLS	7.7E+18	7.0E+18	7.7E+18	7.4E+42	1.000	-5.3E+20	5.5E+20	1.000
Bayesian	7.7E+18	7.0E+18	7.7E+18	7.4E+42	1.000	-5.3E+20	5.5E+20	1.000
Cauchy								
LTS	2.000	0.000	0.000	0.000	0.000	1.996	2.004	1.000
Theil	2.000	0.000	0.000	0.000	0.000	1.999	2.001	1.000
Bayesian	1.940	0.026	-0.060	105.108	1.000	-0.094	3.974	1.000
OLS	1.940	0.026	-0.060	105.110	1.000	-0.094	3.974	1.000
CONTAMINATIONS MODEL								
Method	**β**	**MCSE**	**Bias**	**MSE**	**RelMSE**	**LCL**	**UCL**	**Cov. Pr.**
Normal								
OLS	2.019	0.000	0.019	0.000	1.000	2.018	2.020	1.000
LTS	2.017	0.000	0.017	0.000	0.778	2.016	2.018	1.000
Theil	2.016	0.000	0.016	0.000	0.701	2.015	2.017	1.000
Bayesian	2.019	0.000	0.019	0.000	0.998	2.018	2.020	1.000
Lognormal								
OLS	2.003	0.000	0.003	0.000	1.000	2.002	2.005	1.000
LTS	2.005	0.000	0.005	0.000	0.818	2.004	2.006	1.000
Theil	2.004	0.000	0.004	0.000	0.486	2.004	2.005	1.000
Bayesian	2.003	0.000	0.003	0.000	0.997	2.002	2.005	1.000
Cauchy								
OLS	1.940	0.026	-0.060	105.110	1.000	-0.094	3.974	1.000
LTS	2.000	0.000	0.000	0.000	0.000	1.996	2.004	1.000
Theil	2.000	0.000	0.000	0.000	0.000	1.999	2.001	1.000
Bayesian	1.940	0.026	-0.060	105.108	1.000	-0.094	3.974	1.000

α								
n = 100	**MIXTURES MODEL**							
Method	**α**	**MCSE**	**Bias**	**MSE**	**RelMSE**	**LCL**	**UCL**	**Cov. Pr.**
Normal								
LTS	3.818	0.001	-1.182	1.652	0.200	3.718	3.918	1.000
Theil	3.860	0.002	-1.140	1.886	0.200	3.708	4.012	1.000
OLS	2.152	0.002	-2.848	8.578	1.000	2.016	2.288	1.000
Bayesian	1.084	0.002	-3.916	15.807	1.800	0.948	1.221	1.000
Lognormal								
Theil	5.491	0.003	0.491	1.436	0.000	5.274	5.708	1.000
LTS	-3.3E+13	2.7E+13	-3.3E+13	1.1E+32	0.000	-2.1E+15	2.0E+15	0.988
OLS	-2.5E+20	2.3E+20	-2.5E+20	7.8E+45	1.000	-1.8E+22	1.7E+22	0.000
Bayesian	-2.5E+20	2.3E+20	-2.5E+20	7.9E+45	1.000	-1.8E+22	1.7E+22	0.000
Cauchy								
LTS	4.996	0.003	-0.004	1.458	0.000	4.757	5.236	0.974
Theil	5.011	0.006	0.011	5.547	0.000	4.544	5.478	0.999
Bayesian	7.289	1.796	2.289	4.8E+05	1.000	-130.734	145.312	0.726
OLS	8.341	1.801	3.341	4.9E+05	1.000	-130.095	146.777	0.644
CONTAMINATIONS MODEL								
Method	**α**	**MCSE**	**Bias**	**MSE**	**RelMSE**	**LCL**	**UCL**	**Cov. Pr.**
Normal								
Bayesian	3.379	0.001	-1.621	2.674	7.249	3.337	3.421	1.000
Theil	4.187	0.002	-0.813	1.139	3.090	4.050	4.324	1.000
OLS	4.431	0.001	-0.569	0.369	1.000	4.389	4.473	1.000
LTS	4.503	0.001	-0.497	0.300	0.814	4.457	4.549	1.000
Lognormal								
Bayesian	5.484	0.001	0.484	0.411	0.159	5.401	5.567	0.967
LTS	5.989	0.001	0.989	1.055	0.409	5.935	6.044	0.948
Theil	6.425	0.003	1.425	3.069	1.190	6.223	6.627	1.000
OLS	6.549	0.001	1.549	2.578	1.000	6.466	6.633	0.346
Cauchy								
LTS	4.996	0.003	-0.004	1.458	0.000	4.757	5.236	0.974
Theil	5.011	0.006	0.011	5.547	0.000	4.544	5.478	0.999
Bayesian	7.289	1.796	2.289	4.8E+05	0.994	-130.734	145.312	0.726
OLS	8.341	1.801	3.341	4.9E+05	1.000	-130.095	146.777	0.644

Table 4b: General Estimators' Performance Results Chart for Some Regression Estimators (Sample Size N=100)

β								
n = 1000	**STANDARD MODEL**							
Method	**β**	**MCSE**	**Bias**	**MSE**	**RelMSE**	**LCL**	**UCL**	**Cov. Pr.**
Normal								
OLS	2.000	0.000	0.000	0.000	0.000	2.000	2.000	1.000
LTS	2.000	0.000	0.000	0.000	0.000	2.000	2.000	1.000
Theil	2.000	0.000	0.000	0.000	0.000	2.000	2.000	1.000
Bayesian	2.000	0.000	0.000	0.000	0.000	2.000	2.000	1.000
Lognormal								
OLS	2.000	0.000	0.000	0.000	0.000	2.000	2.000	1.000
LTS	2.000	0.000	0.000	0.000	0.000	2.000	2.000	1.000
Theil	2.000	0.000	0.000	0.000	0.000	2.000	2.000	1.000
Bayesian	2.000	0.000	0.000	0.000	0.000	2.000	2.000	1.000
Cauchy								
LTS	2.000	0.000	0.000	0.000	0.000	2.000	2.000	1.000
Theil	2.000	0.000	0.000	0.000	0.000	2.000	2.000	1.000
OLS	1.985	0.022	-0.015	69.504	1.000	1.468	2.502	1.000
Bayesian	1.985	0.022	-0.015	69.504	69.504	1.468	2.502	1.000
OUTLIERS MODEL								
Method	**β**	**MCSE**	**Bias**	**MSE**	**RelMSE**	**LCL**	**UCL**	**Cov. Pr.**
Normal								
Theil	2.000	0.000	0.000	0.000	0.123	2.000	2.000	1.000
LTS	2.000	0.000	0.000	0.000	0.178	2.000	2.000	1.000
OLS	2.000	0.000	0.000	0.000	1.000	2.000	2.000	1.000
Bayesian	2.000	0.000	0.000	0.000	1.000	2.000	2.000	1.000
Lognormal								
Theil	2.000	0.000	0.000	0.000	0.000	2.000	2.000	1.000
LTS	2.0E+07	1.2E+07	2.0E+07	2.3E+19	0.000	-2.8E+08	3.2E+08	1.000
OLS	7.4E+12	2.3E+12	7.4E+12	7.6E+29	1.000	-4.7E+13	6.1E+13	1.000
Bayesian	7.4E+12	2.3E+12	7.4E+12	7.6E+29	1.000	-4.7E+13	6.1E+13	1.000
Cauchy								
LTS	2.000	0.000	0.000	0.000	0.000	2.000	2.000	1.000
Theil	2.000	0.000	0.000	0.000	0.000	2.000	2.000	1.000
OLS	1.985	0.022	-0.015	69.504	1.000	1.468	2.502	1.000
Bayesian	1.985	0.022	-0.015	69.504	1.000	1.468	2.502	1.000

α								
n = 1000	**STANDARD MODEL**							
Method	**α**	**MCSE**	**Bias**	**MSE**	**RelMSE**	**LCL**	**UCL**	**Cov. Pr.**
Normal								
OLS	5.000	0.000	0.000	0.004	1.000	4.996	5.004	1.000
LTS	5.000	0.000	0.000	0.005	1.400	4.995	5.005	1.000
Theil	4.999	0.002	-0.001	0.449	112.100	4.958	5.041	1.000
Bayesian	3.995	0.000	-1.005	1.014	253.500	3.991	3.999	1.000
Lognormal								
Bayesian	5.642	0.000	0.642	0.431	0.200	5.633	5.650	0.986
LTS	6.119	0.000	1.119	1.266	0.500	6.112	6.126	0.421
OLS	6.649	0.000	1.649	2.736	1.000	6.640	6.657	0.000
Theil	6.648	0.003	1.648	3.741	1.400	6.585	6.711	1.000
Cauchy								
LTS	5.004	0.002	0.004	0.741	0.000	4.950	5.057	0.939
OLS	20.096	18.737	15.096	5.3E+07	1.000	-430.212	470.404	0.590
Theil	5.003	0.007	0.003	8.320	8.300	4.824	5.182	1.000
Bayesian	19.084	18.729	14.084	5.3E+07	5.3E+07	-431.032	469.199	0.690
OUTLIERS MODEL								
Method	**α**	**MCSE**	**Bias**	**MSE**	**RelMSE**	**LCL**	**UCL**	**Cov. Pr.**
Normal								
LTS	5.000	0.000	0.000	0.011	0.200	4.993	5.006	1.000
OLS	5.000	0.001	0.000	0.045	1.000	4.986	5.013	1.000
Theil	4.999	0.002	-0.001	0.458	10.100	4.957	5.041	1.000
Bayesian	3.995	0.001	-1.005	1.056	23.300	3.981	4.008	1.000
Lognormal								
Theil	6.642	0.003	1.642	3.724	0.000	6.579	6.705	1.000
LTS	-5.8E+09	3.7E+09	-5.8E+09	2.1E+24	0.000	-9.6E+10	8.4E+10	0.994
OLS	-2.2E+15	6.9E+14	-2.2E+15	7.2E+34	1.000	-1.9E+16	1.4E+16	0.000
Bayesian	-2.2E+15	6.9E+14	-2.2E+15	7.2E+34	1.000	-1.9E+16	1.4E+16	0.000
Cauchy								
LTS	5.004	0.002	0.004	0.741	0.000	4.950	5.057	0.939
Theil	5.003	0.007	0.003	8.320	0.000	4.824	5.182	1.000
Bayesian	19.084	18.729	14.084	5.3E+07	1.000	-431.032	469.199	0.690
OLS	20.096	18.737	15.096	5.3E+07	1.000	-430.212	470.404	0.590

Table 5a: General Estimators’ Performance Results Chart for Some Regression Estimators (Sample Size N=1000)

β								
n = 1000	**MIXTURES MODEL**							
Method	**β**	**MCSE**	**Bias**	**MSE**	**RelMSE**	**LCL**	**UCL**	**Cov. Pr.**
Normal								
Theil	2.002	0.000	0.002	0.000	0.054	2.002	2.002	1.000
LTS	2.004	0.000	0.004	0.000	0.188	2.004	2.004	1.000
OLS	2.010	0.000	0.010	0.000	1.000	2.010	2.010	1.000
Bayesian	2.010	0.000	0.010	0.000	1.000	2.010	2.010	1.000
Lognormal								
Theil	2.002	0.000	0.002	0.000	0.000	2.002	2.002	1.000
LTS	4.3E+11	2.7E+11	4.3E+11	1.1E+28	0.000	-6.2E+12	7.0E+12	1.000
OLS	1.6E+17	5.0E+16	1.6E+17	3.7E+38	1.000	-1.0E+18	1.4E+18	1.000
Bayesian	1.6E+17	5.0E+16	1.6E+17	3.7E+38	1.000	-1.0E+18	1.4E+18	1.000
Cauchy								
LTS	2.000	0.000	0.000	0.000	0.000	2.000	2.000	1.000
Theil	2.000	0.000	0.000	0.000	0.000	2.000	2.000	1.000
OLS	1.985	0.022	-0.015	69.504	1.000	1.468	2.502	1.000
Bayesian	1.985	0.022	-0.015	69.504	1.000	1.468	2.502	1.000
CONTAMINATIONS MODEL								
Method	**β**	**MCSE**	**Bias**	**MSE**	**RelMSE**	**LCL**	**UCL**	**Cov. Pr.**
Normal								
Theil	2.002	0.000	0.002	0.000	0.693	2.002	2.002	1.000
LTS	2.002	0.000	0.002	0.000	0.739	2.002	2.002	1.000
OLS	2.002	0.000	0.002	0.000	1.000	2.002	2.002	1.000
Bayesian	2.002	0.000	0.002	0.000	1.000	2.002	2.002	1.000
Lognormal								
OLS	2.000	0.000	0.000	0.000	1.000	2.000	2.000	1.000
Bayesian	2.000	0.000	0.000	0.000	1.000	2.000	2.000	1.000
Theil	2.000	0.000	0.000	0.000	1.242	2.000	2.000	1.000
LTS	2.001	0.000	0.001	0.000	2.091	2.001	2.001	1.000
Cauchy								
LTS	2.000	0.000	0.000	0.000	0.000	2.000	2.000	1.000
Theil	2.000	0.000	0.000	0.000	0.000	2.000	2.000	1.000
OLS	1.985	0.022	-0.015	69.504	1.000	1.468	2.502	1.000
Bayesian	1.985	0.022	-0.015	69.504	1.000	1.468	2.502	1.000

α								
n = 1000	**MIXTURES MODEL**							
Method	**α**	**MCSE**	**Bias**	**MSE**	**RelMSE**	**LCL**	**UCL**	**Cov. Pr.**
Normal								
LTS	3.847	0.000	-1.153	1.353	0.200	3.838	3.857	1.000
Theil	3.884	0.002	-1.116	1.703	0.200	3.842	3.926	1.000
OLS	2.195	0.001	-2.805	7.914	1.000	2.182	2.208	1.000
Bayesian	1.188	0.001	-3.812	14.578	1.800	1.175	1.201	1.000
Lognormal								
Theil	5.564	0.003	0.564	1.357	0.000	5.501	5.627	1.000
LTS	-1.3E+14	8.2E+13	-1.3E+14	1.0E+33	0.000	-2.1E+15	1.9E+15	0.994
OLS	-4.9E+19	1.5E+19	-4.9E+19	3.5E+43	1.000	-4.2E+20	3.2E+20	0.000
Bayesian	-4.9E+19	1.5E+19	-4.9E+19	3.5E+43	1.000	-4.2E+20	3.2E+20	0.000
Cauchy								
LTS	5.004	0.002	0.004	0.741	0.000	4.950	5.057	0.939
Theil	5.003	0.007	0.003	8.320	0.000	4.824	5.182	1.000
Bayesian	19.084	18.729	14.084	5.3E+07	1.000	-431.032	469.199	0.690
OLS	20.096	18.737	15.096	5.3E+07	1.000	-430.212	470.404	0.590
CONTAMINATIONS MODEL								
Method	**α**	**MCSE**	**Bias**	**MSE**	**RelMSE**	**LCL**	**UCL**	**Cov. Pr.**
Normal								
LTS	4.523	0.000	-0.477	0.233	0.730	4.518	4.527	1.000
OLS	4.439	0.000	-0.561	0.319	1.000	4.435	4.443	1.000
Theil	4.199	0.002	-0.801	1.089	3.414	4.158	4.241	1.000
Bayesian	3.434	0.000	-1.566	2.458	7.702	3.429	3.438	1.000
Lognormal								
Bayesian	5.543	0.000	0.543	0.313	0.129	5.535	5.551	0.997
LTS	5.961	0.000	0.961	0.932	0.385	5.956	5.967	0.893
OLS	6.550	0.000	1.550	2.420	1.000	6.542	6.558	0.000
Theil	6.431	0.003	1.431	3.074	1.271	6.368	6.494	1.000
Cauchy								
LTS	5.004	0.002	0.004	0.741	0.000	4.950	5.057	0.939
Theil	5.003	0.007	0.003	8.320	0.000	4.824	5.182	1.000
Bayesian	19.084	18.729	14.084	5.3E+07	0.999	-431.032	469.199	0.690
OLS	20.096	18.737	15.096	5.3E+07	1.000	-430.212	470.404	0.590

Table 5b: General Estimators' Performance Results Chart for Some Regression Estimators (Sample Size n=1000)